

Design of low-noise transimpedance amplifiers with capacitive feedback

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Abstract This paper reports on a new topology and design methodology for ultra-low noise and high-gain transimpedance amplifiers. This paper also reports on measurement results of two implemented ICs based on the proposed topology and fabricated in 130 nm IBM and 180 nm TSMC technologies. A capacitive feedback topology is implemented as a noise-efficient feedback network, analytical noise calculations in this family of TIA circuits are presented, and optimum noise criterion is derived. The saturation and instability problem of TIA circuits resulted from DC dark current of the input photodiodes is also addressed and a simple yet efficient feedback solution is proposed. The measurement results of 130 nm chip show average input referred noise of $2.67 \text{ pA}/\sqrt{\text{Hz}}$ with bandwidth of 81 kHz to 1.76 GHz and transimpedance gain of $76 \text{ dB}\Omega$ while dissipating 13.7 mW from a 1.5 V power supply, including the output buffer. The measurement results of 180 nm chip show average input referred noise of $3.18 \text{ pA}/\sqrt{\text{Hz}}$ with bandwidth of 72 kHz to 1.62 GHz and transimpedance gain of $75 \text{ dB}\Omega$ while dissipating 26.3 mW from a 2.2 V power supply, including the output buffer.

Keywords Transimpedance amplifier · Optical communication circuits · Low-noise and high-gain amplifiers · Capacitive feedback

1 Introduction

Transimpedance amplifiers have gained popularity in a wide range of applications such as optical communications (families of 2.5, 10, 25, ... Gb/s), mechanical sensors, biosensors, other bio-specific applications like DNA sequencing, and impedance spectroscopy [1–5]. Each of these versatile applications impose their own set of constraints as “required characterization for the TIAs” to be suitable for their specific goals. While in high speed families of optical communication circuits, the main goal is pushing the bandwidth of TIA higher in GHz range [5] bio-application TIAs usually stay in much lower frequency ranges, still demanding better noise performances and lower power consumption for chronic/long-term implantable versions [2, 4].

Consequently a key issue to be addressed in TIA design is which application the TIA is being used for and what the specific design challenges are for that specific application. Still, there are some general characteristics required for a TIA. As the first building block in a receiving front end, regardless of application, TIAs convert the input current to an amplified voltage to be used in the following blocks. Overall system performance in terms of bandwidth, sensitivity and signal to noise ratio (SNR) is strongly influenced by the first block. Hence, having enough bandwidth for the application, high and precise transimpedance gain in the bandwidth of interest, and low input noise current for higher sensitivity while dissipating reasonable power are the fundamental features expected from a well-designed TIA.

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In TIA design for 2.5 Gb/s optical communications family, the fact that the bandwidth is still in range of few GHz, leads to higher gain and lower noise, at the same time, to be the most important design challenges. Increasing gain usually increases chance of instability and low noise performance usually translates into higher power consumption. Especially in terms of instability, in addition to typical issues that designers should take into account for a voltage amplifier, such as phase margin and gain margin, saturation and instability resulting from DC component of the input current can add an extra level of difficulty for TIA design [5]. The input current is not ideally a symmetric AC current at the input of TIA and usually there is a non-ideal DC current component. In optical communication circuits this DC current, so called “dark” current, comes from mismatch between the two inverse-biased series input photodiodes (PDs) that are used to sense the optical beams at the end of optical fiber. For each of the mentioned problems, understanding the problem and fundamental limitations is the key issue which can result in finding an optimum solution.

2 General TIA topology

General accepted topology for TIA circuit is a voltage amplifier with shunt–shunt resistive feedback across it [5–8], as shown in Fig. 1. This feedback topology is opted because of low input impedance, as desired for a current-input block and low output impedance, as needed for a voltage-output amplifier. This resistive feedback network also assures almost constant transimpedance gain in the bandwidth of interest, while decreasing sensitivity to process and temperature variation.

In this section via a step by step procedure, this general topology is analysed and fundamental limitations and drawbacks will be addressed. Based on the provided analysis, a modified topology will be presented for superior noise and gain performance.

While the noise performance of the TIA is an important design challenge, the simpler voltage amplifier with less

number of active components is opted, the less input current-noise will be resulted. So, a single-stage amplifier can be a viable candidate if enough gain can be achieved. Besides complexity, potentially large inherent input parasitic capacitance in TIAs is another key degrading factor in terms of noise and bandwidth performance. This capacitance (C_{in}) consists of the parasitic capacitance of photodiodes, input pads, and electro-static discharge (ESD) circuit capacitance.

So a single stage voltage amplifier with low input capacitance is desirable choice at this point. Different input topologies have been tested to relax the effect of input capacitance; common gate (CG) [9], regulated cascode (RGC) [10], common drain (CD) [11], and common source (CS) [12]. At first glance, CG and RGC topologies seem better candidates in terms of relaxing the effect of input parasitic capacitances, but high current noise contributions of the load and input transistors in these topologies, are major drawback in terms of noise performance [5]. Although isolating property of CD topology can be useful, but it still needs another stage for voltage amplification. Besides, high noise contribution of the buffer transistor in addition to the amplified (divided by smaller than unity gain of buffer) noise from second gain stage makes this topology a poor candidate in terms of noise efficiency. Among all those mentioned topologies, common source stage can provide enough gain without sacrificing reasonable isolating property.

The most common approach for choice of feedback network as mentioned earlier is a resistive feedback network, as shown in Fig. 1. Resistor value doesn't change in the bandwidth of interest and also, theoretically, provides DC current path for the dark current of PDs. Still, in most cases because of headroom limitations another DC path is designed and implemented and this resistor doesn't serve as a DC path for input current. However, the main drawback of resistive feedback stems from its noise performance. Current noise of this resistor is directly added to the input referred current noise of the TIA, regardless of choice of input stage, which degrades the noise performance drastically [5]. A capacitive noiseless feedback network can serve as a viable solution for this problem. As shown in Fig. 2, we proposed a TIA with capacitive feedback network for maintaining high gain and low noise at the same time [12]. Also a circuit level solution for the DC dark current in this topology is provided.

3 Principles of the proposed topology

In this section the foundations of the proposed topology will be introduced. Then basic characteristics of the proposed topology such as bandwidth, noise, and

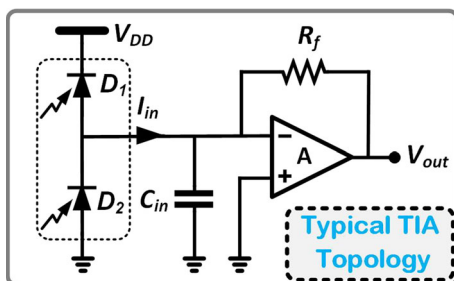


Fig. 1 A transimpedance amplifier with a core voltage amplifier and a resistive shunt–shunt feedback

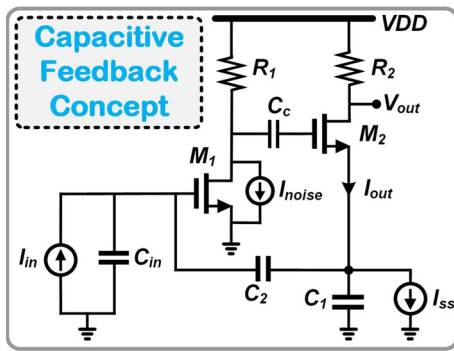


Fig. 2 Proposed TIA topology concept with common source core voltage amplifier and capacitive feedback

transimpedance gain are analysed. As shown in Fig. 2 a capacitive network (including C_1 and C_2) is substituted with the resistive feedback. Also a common source voltage amplifier is used as the core voltage amplifier. In this topology input current is amplified and then is converted to voltage by R_2 resistor ($\frac{V_{out}}{I_{in}} = \frac{I_{out}}{I_{in}} \times R_2$).

In this figure I_{in} represents the input current from photodiodes and C_{in} is the sum of all capacitances at the input node discussed earlier. The transfer function of the output voltage to the input current is shown in Eq. (1).

$$TI \text{ Gain} = \frac{V_{out}}{I_{in}} = \frac{R_2 \times \frac{(g_{m1}R_1(C_1+C_2)+C_2)}{(C_{in}+C_2+g_{m1}R_1C_2)}}{1 + \frac{s(C_1C_{in}+C_1C_2+C_2C_{in})}{g_{m2}(C_{in}+C_2+g_{m1}R_1C_2)}} \quad (1)$$

In this equation, g_m represents the transconductance of corresponding transistors and $g_{m1}R_1$ is the ideal gain from common source amplifier stage. Assuming $g_{m1}R_1 \gg 1$, the current gain in low frequencies can be approximated by $R_2 \times (1 + C_1/C_2)$. As can be seen by the formula, the ratio of the capacitors defines the feedback gain, and that is why they can be implemented by poly capacitors to be more area efficient. On the other hand C_c capacitor should be implemented by a metal insulator metal (MIM) capacitor as it is in the signal path.

The -3 dB bandwidth of the circuit can be approximated with Eq. (2).

$$f_{-3dB} \approx \frac{g_{m2} \cdot (1 + g_{m1}R_1)}{2\pi \cdot C_1} \quad (2)$$

As expected, increasing the transconductance will increase the bandwidth at the cost of consuming more power. Besides, C_1 is the main limiting factor for the bandwidth, which results in opting for a small C_2 capacitor to have the maximum gain while maintaining required bandwidth. Still, for optimum matching, C_2 is chosen to be five times the preferred minimum-size capacitor in each design.

Regarding input referred current noise, main contributors should be addressed. The capacitive feedback network

is noiseless and contributes no noise as opposed to the traditional resistive feedback network. Besides, current thermal noise of M_2 and R_2 will be suppressed when divided by the current gain of the circuit. Therefore the thermal current noise of transistor M_1 and R_1 are the dominant noise sources in this TIA topology. As shown in Fig. 2, I_{noise} represents the thermal noise of M_1 and R_1 . The transfer function from I_{noise} to the output node is shown in Eq. (3).

$$\frac{V_{out}}{I_{noise}} = \frac{\frac{R_1R_2(C_1C_{in}+C_1C_2+C_2C_{in})}{C_{in}+C_2+g_{m1}R_1C_2}}{1 + s \frac{C_1C_{in}+C_1C_2+C_2C_{in}}{g_{m2}(C_{in}+C_2+g_{m1}R_1C_2)}} \quad (3)$$

where

$$C_{in} = C_{PD} + C_{gs} + (1 + g_{m1}R_1)C_{gd} \quad (4)$$

Dividing the noise transfer function by the transimpedance gain in Eq. (1) will results in input referred noise as shown in Eq. (5).

$$\overline{I_{n,in}^2} = \frac{4KT\gamma}{g_{m1}} \cdot \left(1 + \frac{1}{g_{m1}R_1}\right) \cdot \left[\left(\frac{C_1}{1 + C_1/C_2} + C_{in}\right) \cdot s\right]^2 \quad (5)$$

In which K is the Boltzmann constant, T is the absolute temperature in Kelvin, and γ is the noise excess factor. Assuming $g_{m1}R_1 \gg 1$ we will have:

$$\overline{I_{n,in}^2} = \frac{4KT\gamma}{g_{m1}} \cdot \left[\left(\frac{C_1}{1 + C_1/C_2} + C_{in}\right) \cdot s\right]^2 \quad (6)$$

By using the transconductance formula in Eq. (7)

$$g_m = 2\pi \cdot (f_T \cdot C_{gs}) \quad (7)$$

where f_T is the cut-off frequency. Assuming that M_1 is biased to achieve the maximum achievable f_T of the technology, g_m of M_1 is proportional to its C_{gs} according to Eq. (7). Thus (by writing $C_{in} = C_{gs} + C_{PD} + C_{ESD}$ and ignoring C_{gd}) we will have

$$\overline{I_{n,in}^2} \propto \frac{\left(C_{gs} + C_{PD} + C_{ESD} + \frac{C_1}{1+C_1/C_2}\right)^2}{C_{gs}} \quad (8)$$

Equation (8) can be minimized if the condition in Eq. (9) is satisfied.

$$C_{gs} = C_{PD} + C_{ESD} + \frac{C_1}{1 + C_1/C_2} \quad (9)$$

Equation (9) shows the minimum input current noise criterion in this topology. The following procedure can be used to obtain the component values of the circuit considering the power constraint for each stage. C_1 can be calculated from Eq. (2) as g_{m2} is determined by the power budget of the second stage. C_2 and R_2 can be chosen from the required transimpedance gain in Eq. (1). Finally, the

aspect ratio (W/L) of M_1 can be calculated considering the optimization condition of Eq. (9).

4 Improved circuit topology

In the previous section, the general topology and performance characteristics of a transimpedance amplifier with single stage common source core voltage amplifier and noiseless capacitive feedback was discussed. In this section, the proposed topology will be discussed in further details and its features including noise, bandwidth, and transimpedance gain will be improved at circuit level in a step by step procedure to get to the final implemented circuit while essentially retaining the proposed topology. Besides, a simple yet efficient feedback solution is proposed to mitigate the DC dark current issue in most transimpedance amplifier design which can be used in other topologies as well.

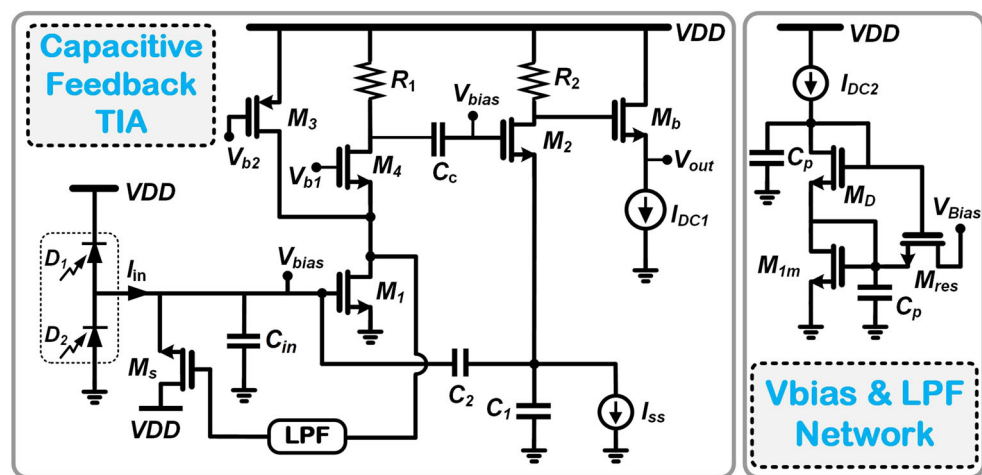
Schematic of the proposed TIA is depicted in Fig. 3. The higher gain the core amplifier provides, the more accurate the feedback gain will be. To achieve this goal, R_1 can be increased but the headroom limitation will be problematic. To increase R_1 without sacrificing the current density in the first stage transistor (M_1), a PMOS transistor (M_3) is added in parallel to its load to handle the current fed into M_1 . By increasing R_1 its noise current is decreased and M_3 transistor is biased optimally (V_{b2}) to have minimum noise current, which ultimately results in a decreased noise contribution from the load. Reducing the input capacitance can also play an important role in improving noise efficiency. Specifically, the Miller effect of gate-drain capacitance of M_1 can be reduced by stacking an NMOS transistor (M_4) on top of M_1 .

In optical communications, the low cut-off frequency of 50 kHz is required in most of optical communication

standards [5]. The lower end of the amplifier bandwidth is critical because it determines how much of the energy of the random binary signal is removed and how much intersymbol interface (ISI) is generated, which ultimately controls the eye closure in eye diagram. The lower low-end cut-off frequency is achieved, the less eye closure will result. To achieve such low-end cut-off frequencies, a very large resistor should be connected in series with the gate of M_1 and M_2 , with minimal parasitic capacitances. Implementing such large resistors (few M Ω) will occupy a large silicon area and will have large parasitic capacitances. Instead, a three-transistor topology depicted in Fig. 3 (V_{bias} generator) is implemented to bias the gate of M_1 and M_2 . In this network, I_{DC2} and M_D define the on resistance of M_{res} and M_{1m} and M_1 constitute a current mirror. If $(W/L)_{MD} = 100 \times (W/L)_{Mres}$, then $R_{onMres} = 100 g_m^{-1} MD$. This resistor occupies considerably less silicon area as compared to using regular available resistors in CMOS processes, and constitutes a low pass filter (LPF) network with the parasitic capacitances (C_p) shown in the Fig. 3.

To address the DC dark current, that leads to instability and saturation problems in transimpedance amplifiers, instead of using off-chip solutions [10], we proposed an efficient solution. As shown in Fig. 3, pair of NMOS transistors, M_s and M_{1m} , have been implemented at the gate of input transistor, M_1 . In case of extra DC current, sourcing to the gate of M_1 , diode connected transistor M_{1m} will sink the extra current to the ground, hence limiting the maximum voltage at the gate of M_1 . In case of extra DC current sinking from the input node, transistor M_s will turn on and will provide the extra current needed for the required period of time. Transistor M_s will be off in the absence of extra sinking current as long as the inequality in Eq. (10) is satisfied. It will turn on only if the voltage at the gate of M_1 drops and voltage at the drain is increased at the same time so that the Eq. (10) does not hold anymore.

Fig. 3 Schematic of the proposed TIA including core common source voltage amplifier, capacitive feedback network, feedback circuit to eliminate DC component, and V_{bias} generator network which incorporates a LPF. Bias circuits, V_{b1} and V_{b2} , are not shown



Transistor M_s is connected to the gate of M_1 through a low pass filter network which incorporated a similar topology as described for V_{bias} network in Fig. 3.

$$V_{G1} - V_{th,M1} < V_{D1} < V_{G1} + V_{th,M_s} \quad (10)$$

The $NMOS$ transistors are optimally designed with minimum widths to have negligible effect on the input noise current of the circuit. Besides, the short activation period of M_s , results in negligible noise contribution to the overall averaged input noise current of the circuit. By applying DC current of up to 200 μA , the DC bias point of the input transistor has changed less than 6 and 7.2 % in the post-layout simulation of 130 and 180 nm chips, respectively.

To match the output of the transimpedance amplifier to 50 Ω loads, as a requirement for most of the voltage-output amplifiers, a buffer stage is added to the output of the proposed topology (M_b). Transistor M_b is biased to have a transconductance of $g_m^{-1} = 50 \Omega$.

5 Measurement procedure and results

Based on the design methodology discussed in previous sections, two chips in 130 nm IBM and 180 nm TSMC technologies were designed, optimized, and implemented

based on their technology variations. In this section, the measurement procedure and results will be presented for both fabricated chips.

In order to measure specifications of a TIA, including TI-gain and bandwidth, different approaches can be taken. A common approach is measuring the s-parameters of the circuit, like an LNA, and then calculating the TIA's parameters by modified formulas. This approach is widely used because almost all the measurement set-ups are made for s-parameters measurement and new instruments are not needed.

The measured s-parameters of the 180 nm circuit are shown in left side of Fig. 4. It can be seen that S_{22} remains less than -15 dB in the working bandwidth, which translates to good output matching to 50 Ω loads. On the right side of Fig. 4 die microphotograph of the chip is shown.

In Fig. 5 measured s-parameters of the 130 nm chip are presented. Flat S_{21} predicts a flat gain in the bandwidth of interest and again S_{21} is well below -18 dB. On the right side of Fig. 5, layout of the 130 nm chip is presented.

In order to calculate the transimpedance gain of a TIA from the s-parameters, the formula in Eq. (11) can be used.

$$\frac{V_{out}}{I_{in}} = Z_t = \frac{2S_{21}}{(1 - S_{11})(1 - S_{22}) - S_{21}S_{12}} Z_0 \quad (11)$$

Fig. 4 Left Measured s-parameters of 180 nm TSMC TIA. Right Die photograph of the implemented chip

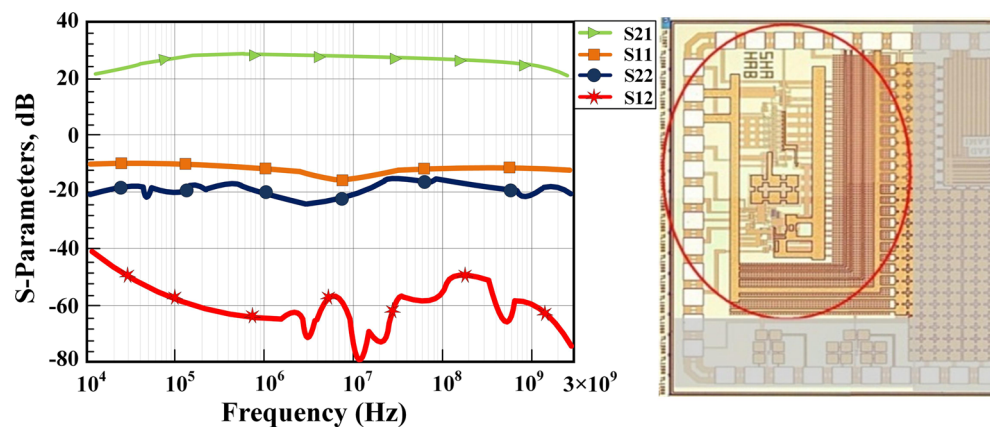


Fig. 5 Left Measured s-parameters of 130 nm IBM TIA. Right Layout of the implemented chip

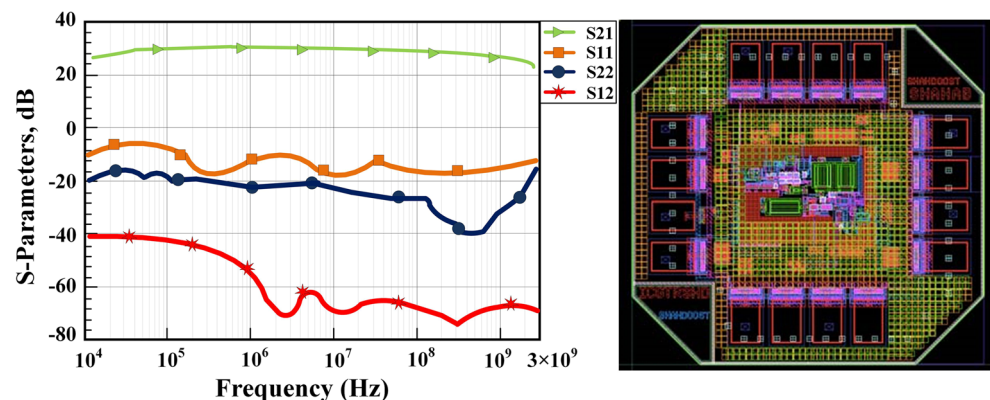


Fig. 6 Measurement versus simulation results of 180 nm TSMC TIA chip. *Left* Transimpedance gain; *right* Input referred current noise

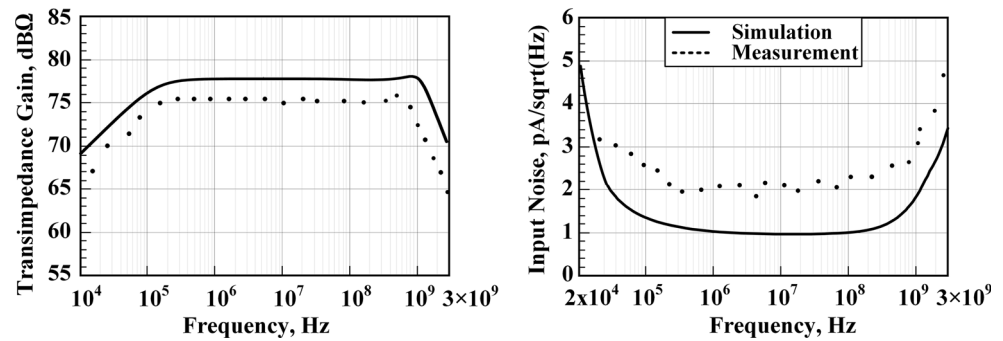


Fig. 7 Measurement versus simulation results of 130 nm IBM TIA chip. *Left* Transimpedance gain; *right* Input referred current noise

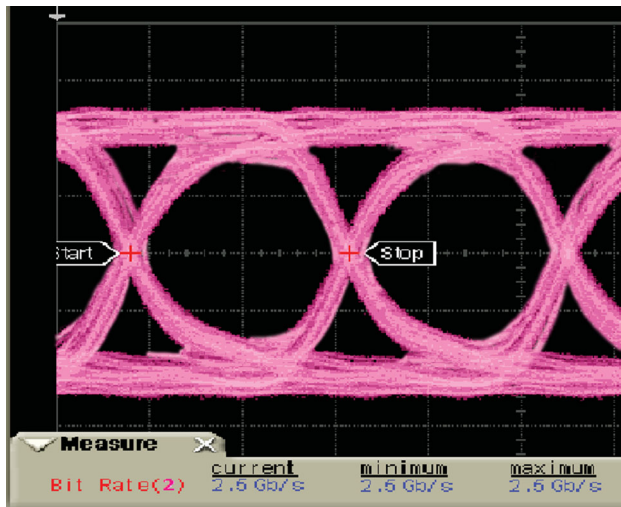
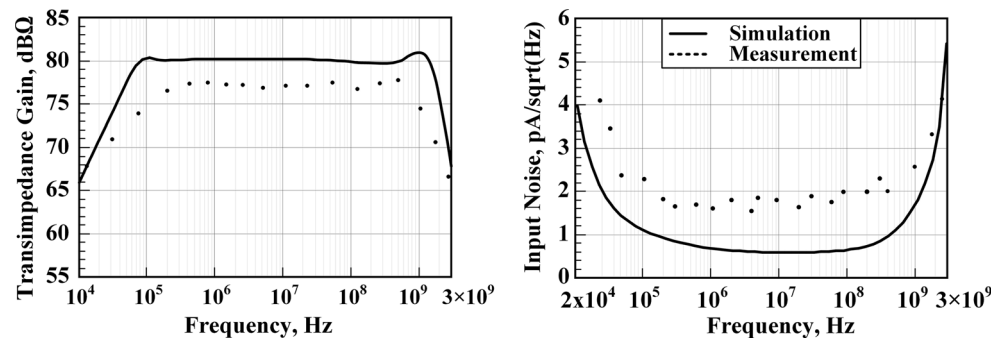


Fig. 8 Measured eye diagram for 180 nm TSMC chip with 100 μ A peak to peak input current of 2.5 Gb/s $2^{31} - 1$ PRSB

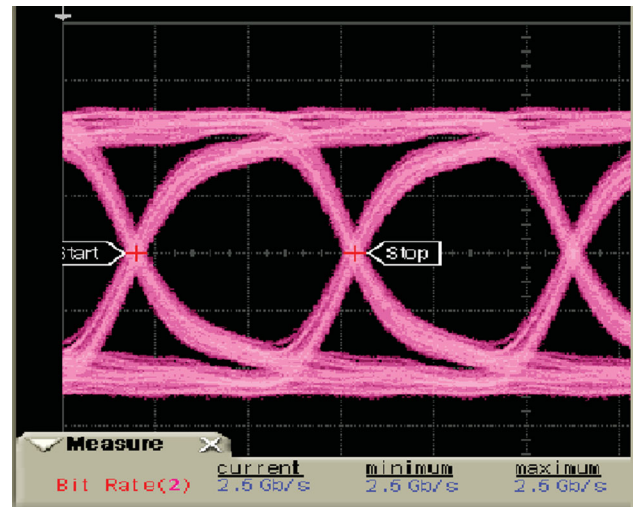


Fig. 9 Measured eye diagram for 130 nm TSMC chip with 100 μ A peak to peak input current of 2.5 Gb/s $2^{31} - 1$ PRSB

In which Z_0 is 50 Ω . This formula can be used only when the input of the TIA is directly connected to the network analyser, and there is no input matching. By using the above formula and the measured s-parameters, the transimpedance gain of both TIA circuits in 180 and 130 nm have been calculated and shown in the following figures.

For 180 nm chip, the post-layout simulated transimpedance gain was a flat gain of 78 dB Ω with -3 dB

bandwidth between 51 kHz and 1.81 GHz, while as it can be seen in Fig. 6, the transimpedance gain in the measurement is 75 dB Ω . Also, measured bandwidth of the circuit is around 1.62 GHz. The peaking in the frequency response is less than 1.3 dB.

For noise measurement, output noise voltage of the TIA was lower than noise floor of the spectrum analyser, therefore extra amplification was needed. After amplifying the output noise, the noise figures of the added stages are

Table 1 Measurement and simulation results of this work compared with results of similar recent works

	This work measured/simulated	This work measured/simulated	[13]	[14]	[15]	[16]	[17]	[18]
Technology CMOS (nm)	130	180	130	180	180	130	40	350
TI gain (dBΩ)	76/80	75/78	53.6	46	55	57	79.5	107
Max BW (GHz)	1.76/1.9	1.62/1.81	6	4	7	8	1.5	0.125
Power (mW)	13.7/12	26.3/26	15.7	10.7 ^a	18.6	1.8	4.5	60
Avg. input current noise (pA/√Hz)	2.67/1.93	3.18/2.1	24	10	17.5	30	8.85	3.54

^a Excluding output buffer

taken into account for noise calculations. Figure 6 also shows the measurement and simulation results of input referred noise current for 180 nm TIA.

The post-layout simulation results predicted input referred noise current of 1.84 pA/√Hz at 1 GHz and an average of 2.1 pA/√Hz within the bandwidth of the circuit. Measurement results shows an average input referred noise current of 3.18 pA/√Hz within the frequency bandwidth of the transimpedance amplifier. Slightly higher noise in the measurement results can be due to parasitic capacitances from PCB and measurement set-up which was added to input capacitance (C_{in}) and as predicted by Eq. (5), extra input capacitance results in higher input noise. This TIA occupies $600 \times 300 \mu\text{m}^2$ of silicon area.

Figure 7 shows the measurement and simulation results for IBM 130 nm TIA. Transimpedance gain versus frequency is depicted on left which shows a flat TI-gain of 80 dBΩ in 52 kHz up to 1.9 GHz simulated BW, while in measurement TI-gain was 76 dBΩ within 81 kHz up to 1.76 GHz of the measured bandwidth for the TIA.

The post-layout simulation results, shown on the right, predicted input referred noise current of 1.74 pA/√Hz at 1 GHz and an average of 1.93 pA/√Hz within the bandwidth of the circuit. Measurement results shows an average input referred noise current of 2.67 pA/√Hz within the frequency bandwidth of the transimpedance amplifier. This TIA occupies $400 \times 300 \mu\text{m}^2$ of silicon area. In both chips, transimpedance gain was close to simulation results and the small drop (less than 3 dB for both chips) can be due to extra parasitic capacitors introduced to the circuit by printed circuit board (PCB) or measurement equipment.

Figure 8 shows the eye diagram for 180 nm chip measured by Agilent 86105C with 2.5 Gb/s, $2^{31} - 1$ PRSB data pattern generated by Agilent J-BERT N4903A. The output has a peak to peak value around 0.495 V with input level to be 100 μA and peak to peak jitter of 24ps. Figure 9 shows the measured eye diagram for the 130 nm IBM technology chip. Input has the same characteristics and the output has a peak to peak value of 0.58 V with input level to be 100 μA and peak to peak jitter of 21ps.

In Table 1, fundamental measured and simulated characteristics of our implemented TIAs in both technologies

are summarized and compared with state of the art TIAs in the literature.

6 Conclusion

This paper presented a thorough design methodology based on a novel topology for high speed and low noise transimpedance amplifier design. The proposed capacitive feedback eliminates the disadvantages of typical resistive feedback while maintaining the feedback characteristics. Besides, in this topology gain is ratio of two capacitors which make it less vulnerable to process variation compared to absolute value of a resistor. Furthermore, a simple yet efficient feedback topology for alleviating the DC dark current problem in TIA design was proposed which can be used in other topologies as well. The proposed design was implemented in two different CMOS processes, namely 130 nm IBM and 180 nm TSMC. Measured gain and bandwidth are reasonably close to simulation results for both chips. Measurement results also showed promising noise performance in both fabricated chips while consuming ultra-low power. These two features combined make the capacitive feedback TIA topology as a viable candidate for low noise applications, especially bio applications, or implantable devices [19, 20] for long-term performances.

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